New York City acquired the Sahara Desert. The total property taxes collected would go up, but by a tiny amount compared with the total increase in acreage.

Ever since the end of inflation, gravity has been trying to make up the entropy difference. Every clump—be it a galaxy, or a star in a galaxy, or a planet, or a black hole—that gravity has subsequently coaxed out of the uniformity (seeded by the tiny nonuniformity from quantum jitters) has increased entropy and has brought gravity one step closer to realizing its entropy potential. In this sense, then, inflation is a mechanism that yielded a large universe with relatively low gravitational entropy, and in that way set the stage for the subsequent billions of years of gravitational clumping whose effects we now witness. And so inflationary cosmology gives a direction to time's arrow by generating a past with exceedingly low gravitational entropy; the future is the direction in which this entropy grows.[4]

The second issue becomes apparent when we continue down the path to which time's arrow led us in Chapter 6. From an egg, to the chicken that laid it, to the chicken's feed, to the plant kingdom, to the sun's heat and light, to the big bang's uniformly distributed primordial gas, we followed the universe's evolution into a past that had ever greater order, at each stage pushing the puzzle of low entropy one step further back in time. We have just now realized that an even earlier stage of inflationary expansion can naturally explain the smooth and uniform aftermath of the bang. But what about inflation itself? Can we explain the initial link in this chain we've followed? Can we explain why conditions were right for an inflationary burst to happen at all?

This is an issue of paramount importance. No matter how many puzzles inflationary cosmology resolves in theory, if an era of inflationary expansion never took place, the approach will be rendered irrelevant. Moreover, since we can't go back to the early universe and determine directly whether inflation occurred, assessing whether we've made real progress in setting a direction to time's arrow requires that we determine the *likelihood* that the conditions necessary for an inflationary burst were achieved. That is, physicists bristle at the standard big bang's reliance on finely tuned homogeneous initial conditions that, while observationally motivated, are theoretically unexplained. It feels deeply unsatisfying for